Protocol for nuclei isolation from mouse lung for single nucleus RNASeq

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Jeffrey R Koenitzer and Benjamin D Humphreys

This protocol is adapted directly from Joshi et al. https://www.protocols.io/view/single-nucleus-isolation-from-frozen-human-lung-ti-zu8f6zw**, with adjustments to RNase inhibitor concentrations and removal of FACS sorting steps.**

Materials	Company	Cat. No.	Storage
Reagent for buffer and stock			
1x DPBS	Gibco	14190144	4 ºC
Nuclei EZ Lysis Buffer (from kit)	Sigma	N 3408 (from NUC-101)	4 ºC
cOmplete ULTRA Tablets, Mini, EDTA-free, EASYpack	Roche	05 892 791 001	4 ºC
RNasin Plus Ribonuclease Inhibitors	Promega	N2615	-20°C
SUPERaseIN RNase Inhibitor	Thermo Fisher Scientific	AM2696	-20°C
RNase free H_2O	Thermo Scientific	AM9938	RT
RNaseZap	Ambion	AM9780	RT
Bovine serum albumin, 10% solution, nuclease free	Millipore Sigma	1187358001	-20°C
Tools			
GentleMacs C tube	Miltenyi Biotech	130-093-237	
GentleMacs Dissociator	Miltenyi Biotech	130-093-235	
pluriStrainer 40 µm	pluriSelect	43-50040	
pluriStrainer 5 µm	pluriSelect	43-50005	
Fuchs-Rosenthal disposable hemocytometer	INCYTO	DHC-F015	
Falcon 15 mL Polystyrene Conical Tube	Fisher Scientific	352095	
RNase-free 50 ml Conical Tubes	Ambion	AM12502	
TPP 60mm Tissue Culture Dishes	MIDSCI	TP93060	

Buffers and Solutions

Working Solutions

cOmplete stock (10x)

- 1 mL Nuclei EZ Prep lysis buffer
- 1 tablet of cOmplete ULTRA tablets

Lysis Buffer - 2 ml per < 6 mm^3 tissue

- 200 µl cOmplete stock
- 1.775 ml of Nuclei EZ Prep lysis buffer
- 12.5 µl of RNasin Plus
- 12.5 µl of SUPERaseIN

Wash Buffer - 4 ml per < 6 mm^3 tissue

- 3.575 ml of dPBS
- 400 µl of 10% BSA
- 12.5 µl of RNasin Plus
- 12.5 µl of SUPERaseIN

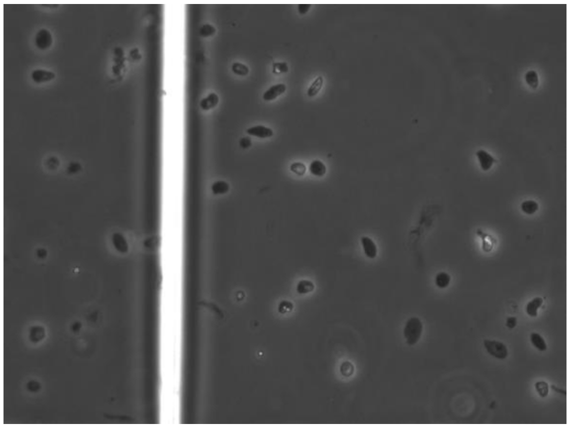
Resuspension Buffer – 1 ml

- 0.987 ml of dPBS
- 10 µl of 10% BSA
- 6.25 µl of RNasin Plus
- 6.25 µl of SUPERaseIN

Procedure

1. Precool all instruments (including centrifuges), buffers, and tubes.
   Note: All steps are performed on ice or in cold room (4ºC) to minimize RNA degradation.
2. Remove mouse lung sample from -80ºC freezer and trim a ~6 mm^3 piece. Thaw on small plastic weighing boat until able to insert 28G needle to tissue, then inject 1 ml of Lysis Buffer to 'inflate' the tissue. Add the remaining Lysis Buffer and chop to the smallest pieces possible with scissors (60 s).
3. Transfer the minced tissue and buffer to a GentleMacs C tube. Close, invert, and transfer directly to MACS Tissue dissociator.
4. Run m_lung_01 program and m_lung_02 program in sequence, stopping the latter after 20 seconds. Place tube on ice.
5. Reduce foam by centrifugation (1 min at 750 x *g*). Use a wide-bore pipette tip to pass lysate through 40 µm strainer to 50 mL conical tube. Wash strainer with 4 mL Wash Buffer.
6. Pass suspension through a 5 µm strainer into 50 ml conical tube.
7. Centrifuge at 600 x *g* for 5 min at 4ºC.
8. Resuspend in 500 µl Resuspension Buffer. Count nuclei by hemocytometer and dilute to desired concentration (e.g. 10,000 per µl). *Note*: it is easier to resuspend in lower volumes and dilute than to concentrate nuclei via further centrifugation.
9. Proceed to 10x Chromium.

Expected Results



40x
Nuclear suspension with scant debris

Foam after GentleMacs (L) and after centrifugation (R)